My Database Theory

1 Difference Between Data and Information

Data refers to raw, unprocessed facts, figures, or symbols that on their own have little or no meaning. For example, the number "42" is a piece of data. Information, on the other hand, is data that has been processed, organised, and given context to provide meaning. If we know that "42" represents the age of a person named John, it becomes information. In essence, data is the input and information is the meaningful output produced after processing that data.

2 What is Metadata?

Metadata is data that describes other data. It provides context and structure that makes raw data easier to understand and interpret. For example, if a database stores temperature readings (17, 25, 20, 24...), the metadata would include information such as the unit of measurement (Celsius), the time period the readings were taken, and the location. Metadata is essential for extracting meaning from data and is a simple but critical form of data processing.

3 What is a DBMS and What Are Its Advantages?

A Database Management System (DBMS) is a collection of programs that manages the database structure and controls access to the data stored in the database. It acts as an intermediary between the user and the database, receiving application requests and translating them into the complex operations needed to fulfil those requests.

Advantages of a DBMS:

- Better data access and security: The DBMS manages database access while enforcing data privacy and security policies, giving users efficient access to data.
- Improved data integration: It facilitates multiple end-users accessing data in a well-managed manner, providing a clearer and more integrated view of an organisation's operations.
- Smooth data manipulation: It enables quick answers to on-the-spot requests for reading or updating data.
- High-quality information: It supports quicker and more informed decision-making, increasing end-user productivit.

4. Operational vs Analytical Databases

Operational Databases (OLTP Online Transaction Processing):
These are designed to support the day-to-day operations of an organisation. They store and manage current, detailed data and are optimised for speed and efficiency in handling frequent updates and retrievals.

Example: An inventory management system that tracks stock levels in real time.

Analytical Databases (OLAP Online Analytical Processing):
These are designed for data analysis and business intelligence. Unlike operational databases, they store historical and aggregated data, optimised for complex queries and reporting to help businesses gain insights into trends and performance.
Example: A data warehouse that stores years of sales data to identify seasonal trends and support strategic planning decisions.

Key Differences:
- Operational databases handle current data; analytical databases handle historical data.
- Operational databases are optimised for transactions; analytical databases are optimised for complex queries and reporting.
- Operational databases support daily operations; analytical databases support decision-making.

5. Types of Data and Use Cases for NoSQL Databases

Databases can be classified by their structure into three types of data:

- Structured data: Highly organised data arranged in a predefined format (rows and columns), as in relational databases.
- Semi-structured data: Has some organisation but lacks the rigidity of structured data. Common formats include XML and JSON.
- Unstructured data: Has no predefined format and includes various types such as text documents, images, and videos

NoSQL databases are most effective in use cases that involve:
- Large volumes of unstructured or semi-structured data (e.g., social media posts, user-generated content)
- Applications that need high scalability and performance under heavy loads (e.g., real-time web applications).
- Scenarios where the data model evolves rapidly and a flexible schema is needed (e.g., content management systems, e-commerce catalogues).
- Big data and real-time analytics where traditional relational databases would struggle with volume and velocity.

6. DBMS Without Server Configuration and Its Advantages

SQLite is the DBMS that does not require server configuration. It is a lightweight, file-based relational database that is self-contained and serverless.

Advantages of SQLite
- No server setup required: It runs entirely within the application, removing the complexity of configuring and maintaining a separate database server.

- Simplicity: Easy to set up and use, making it ideal for beginners and small-scale projects.
- Portability: The entire database is stored in a single file, which can be easily copied, moved, or shared.
- Reliability and speed: Despite its compact size, SQLite offers strong reliability and performance.
- Ideal for embedded systems: It is widely used in mobile applications, embedded systems, and any environment where a lightweight database solution is needed without the overhead of a full server-based DBMS.

7. ACID Properties in the Context of a DBMS

ACID stands for Atomicity, Consistency, Isolation, and Durability. These are the four key properties that relational DBMSs use to ensure data integrity and consistency during database transactions.

- Atomicity: A transaction is treated as a single, indivisible unit. Either all operations within the transaction are completed successfully, or none of them are applied. This prevents partial updates that could corrupt data.

- Consistency: Every transaction brings the database from one valid state to another. The database must always remain in a consistent state before and after a transaction, ensuring data correctness at all times.

- Isolation: Transactions occurring simultaneously (in parallel) are executed independently of one another. The intermediate state of a transaction is not visible to other transactions, preventing conflicts and ensuring data integrity under concurrent access.

- Durability: Once a transaction has been committed, the changes are permanent and will persist even in the event of a system crash or failure. The DBMS uses mechanisms such as logs and backups to ensure durability.

Together, ACID properties guarantee that database transactions are complete, accurate, independent, and permanent, making relational DBMSs reliable for critical data operations.